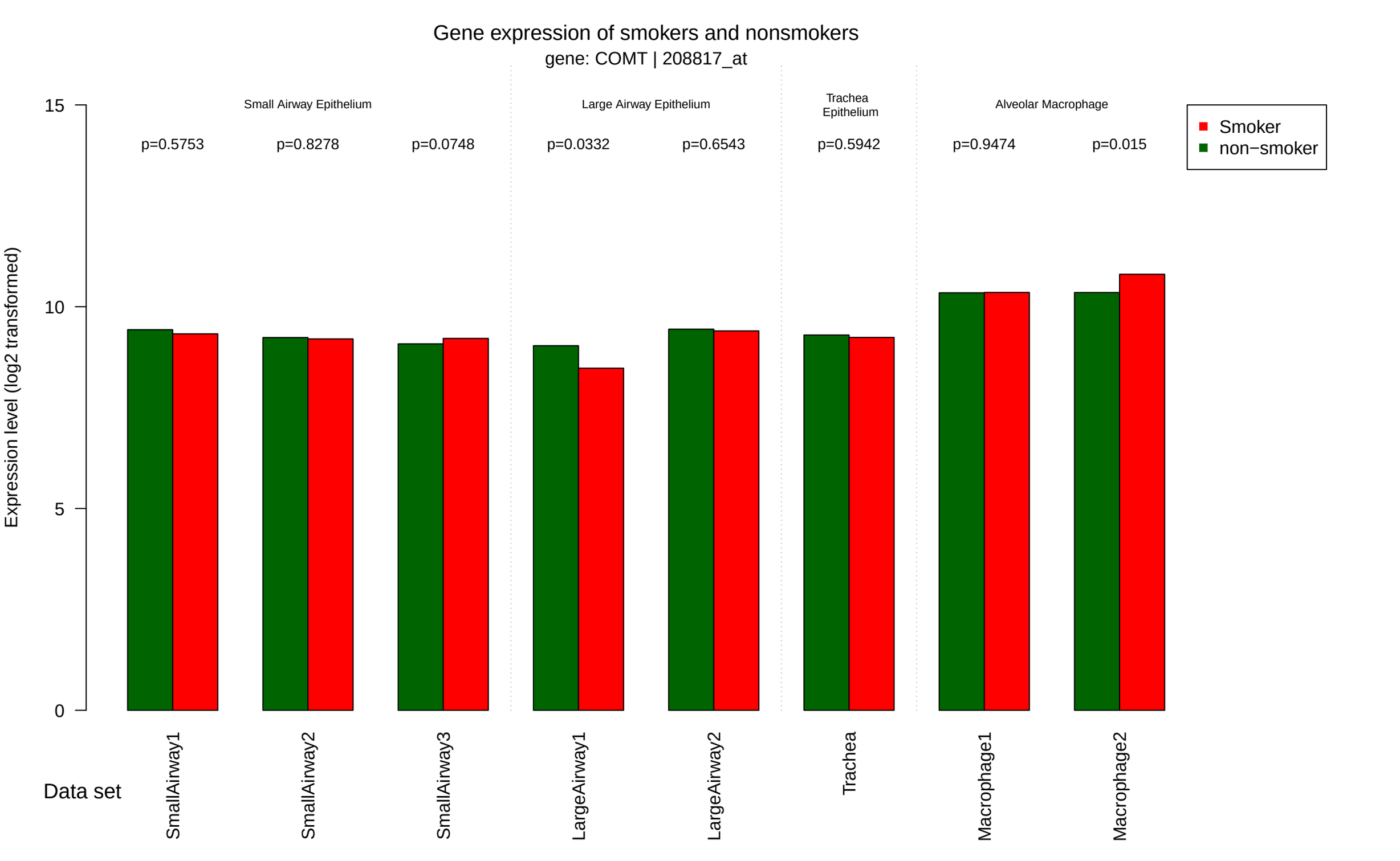
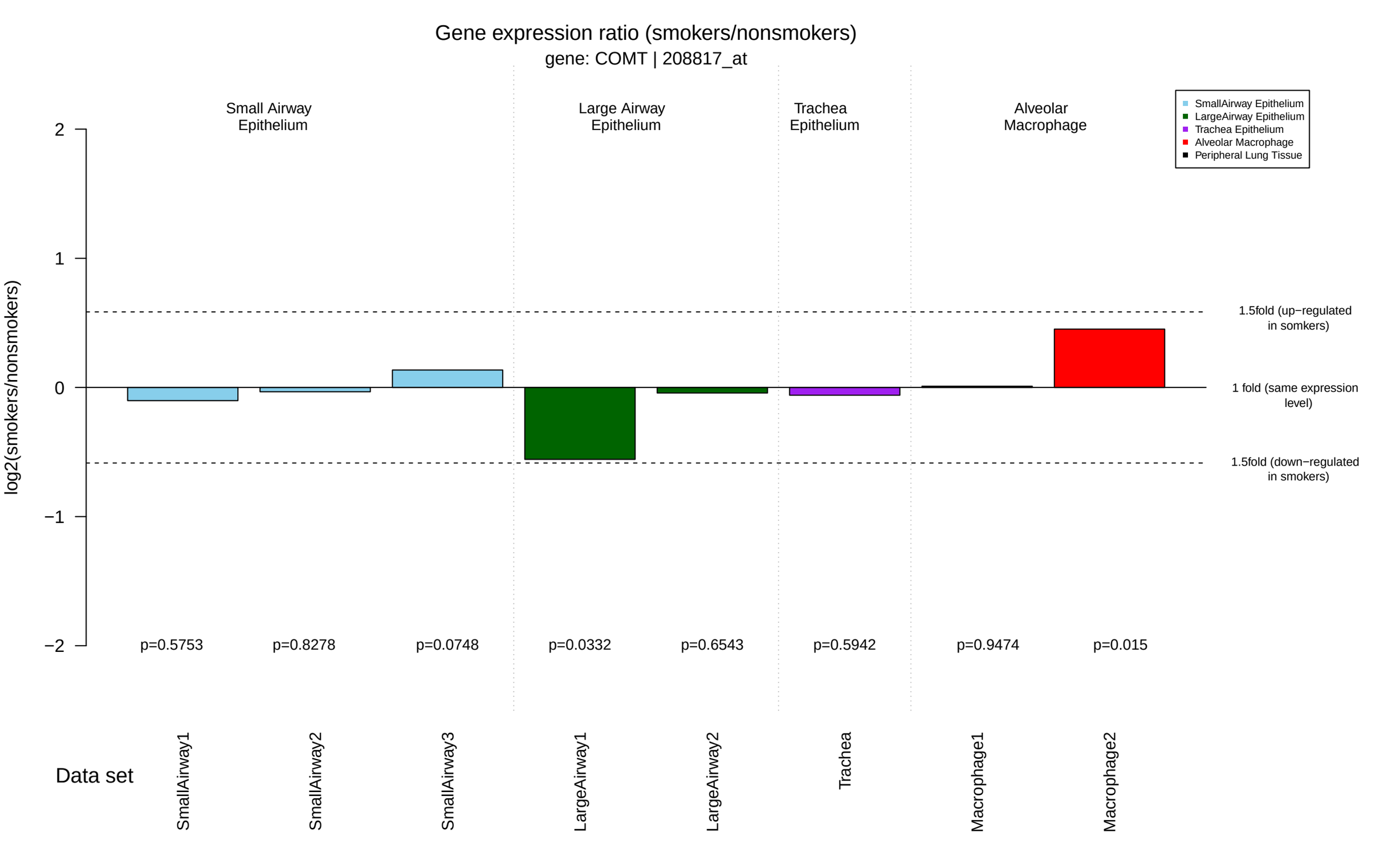
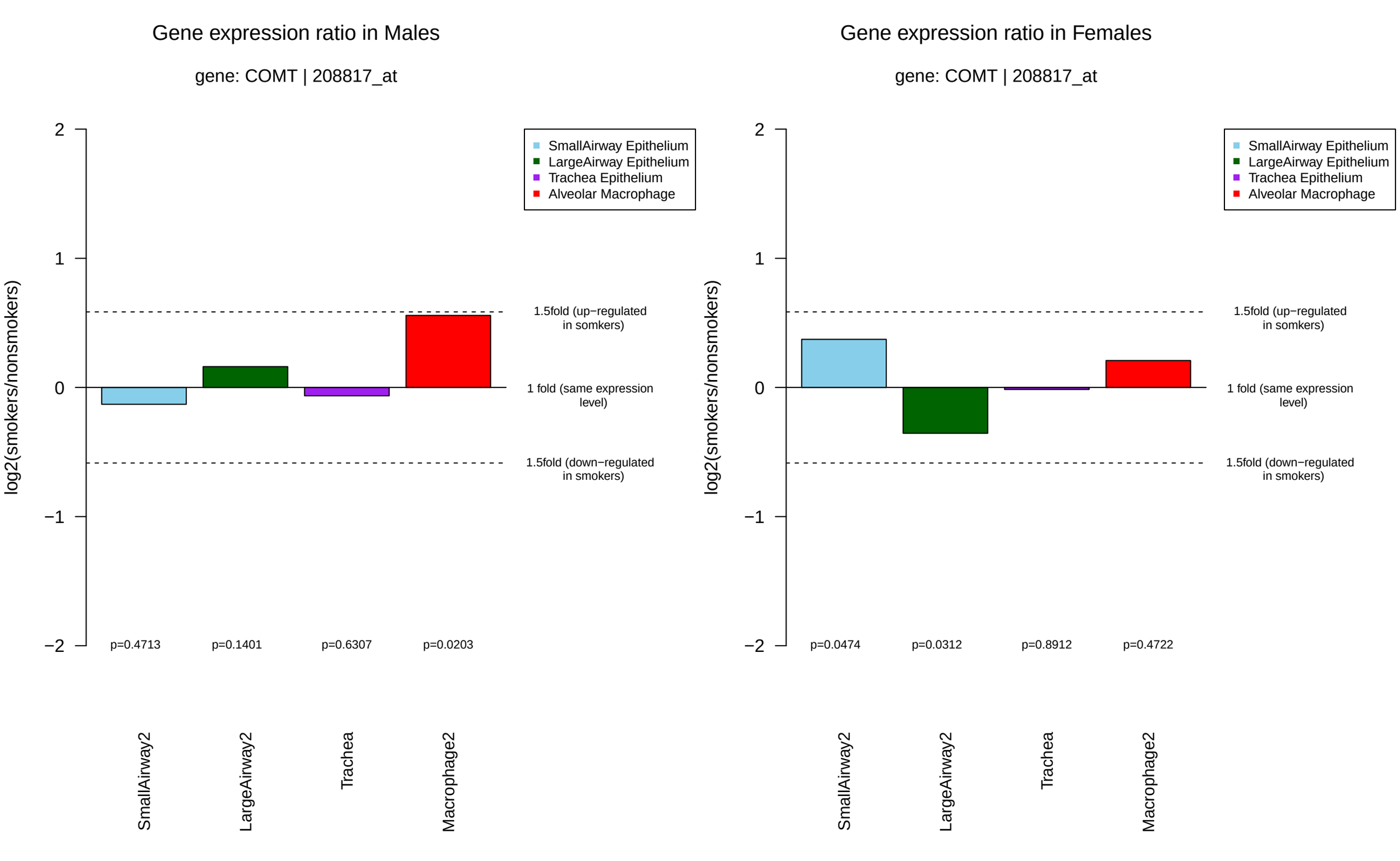
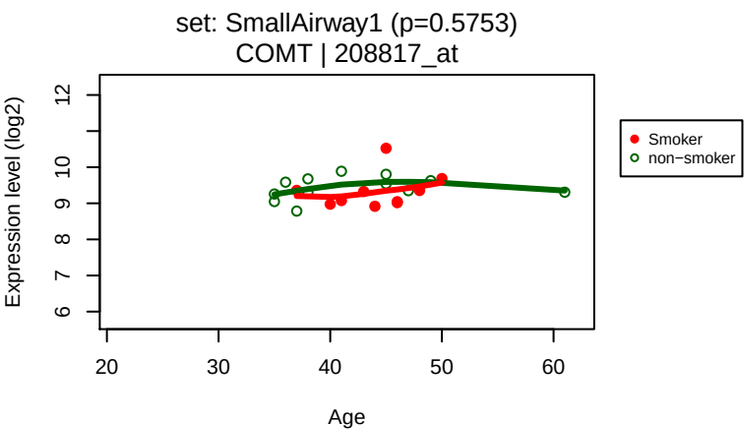
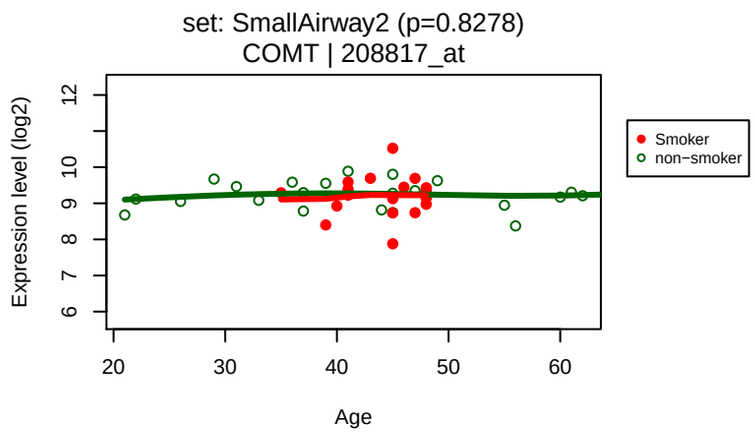
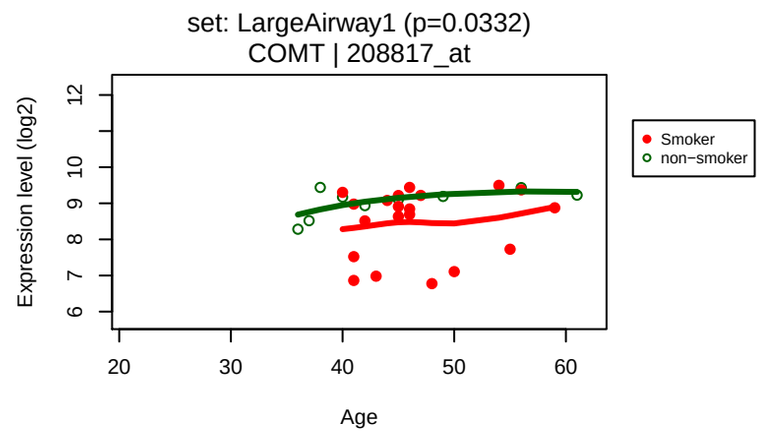

Gene expression of smokers and nonsmokers
gene: COMT | 208817_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5753
p=0.8278
p=0.0748
p=0.0332
p=0.6543
p=0.5942
p=0.9474
p=0.015
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

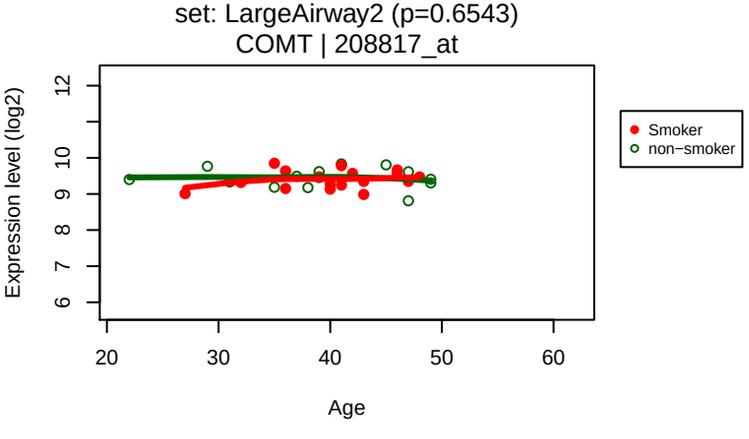

Gene expression ratio (smokers/nonsmokers)
gene: COMT | 208817_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5753
p=0.8278
p=0.0748
p=0.0332
p=0.6543
p=0.5942
p=0.9474
p=0.015
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

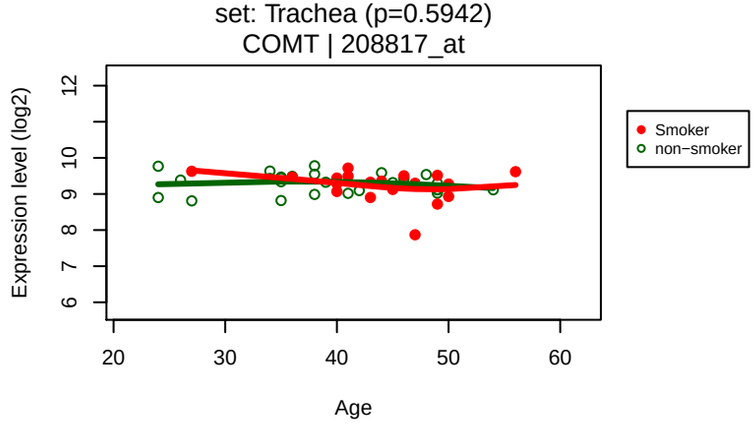

Gene expression ratio in Males
gene: COMT | 208817_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4713
p=0.1401
p=0.6307
p=0.0203
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COMT | 208817_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0474
p=0.0312
p=0.8912
p=0.4722
SmallAirway2
LargeAirway2
Trachea
Macrophage2

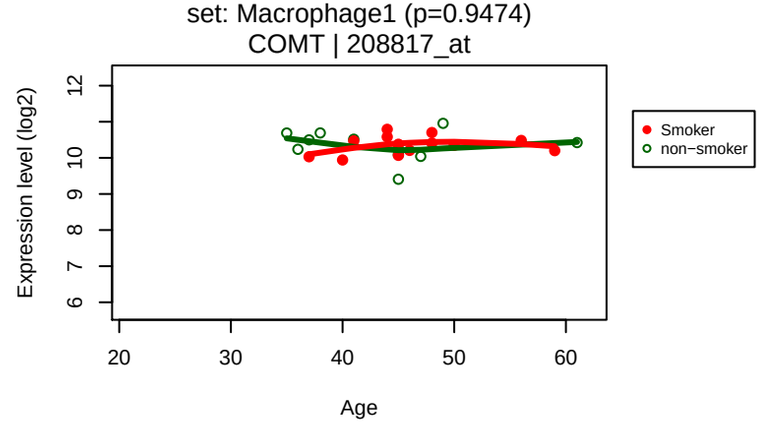

set: SmallAirway1 (p=0.5753)
COMT | 208817_at
Expression level (log2)
Age
Smoker
non−smoker

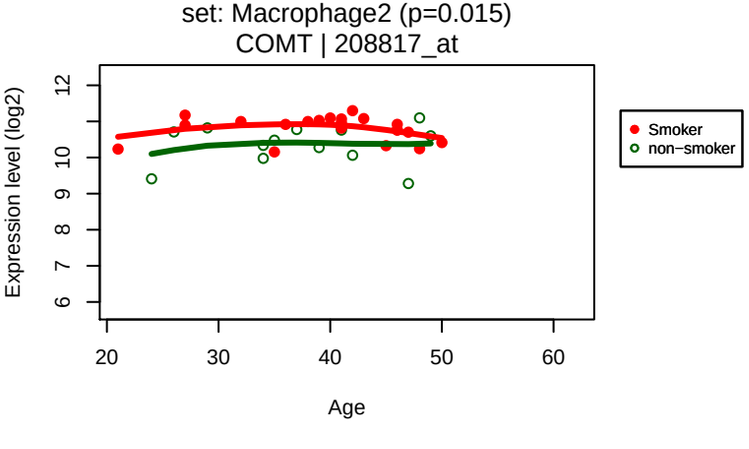

set: SmallAirway2 (p=0.8278)
COMT | 208817_at
Expression level (log2)
Age
Smoker
non−smoker

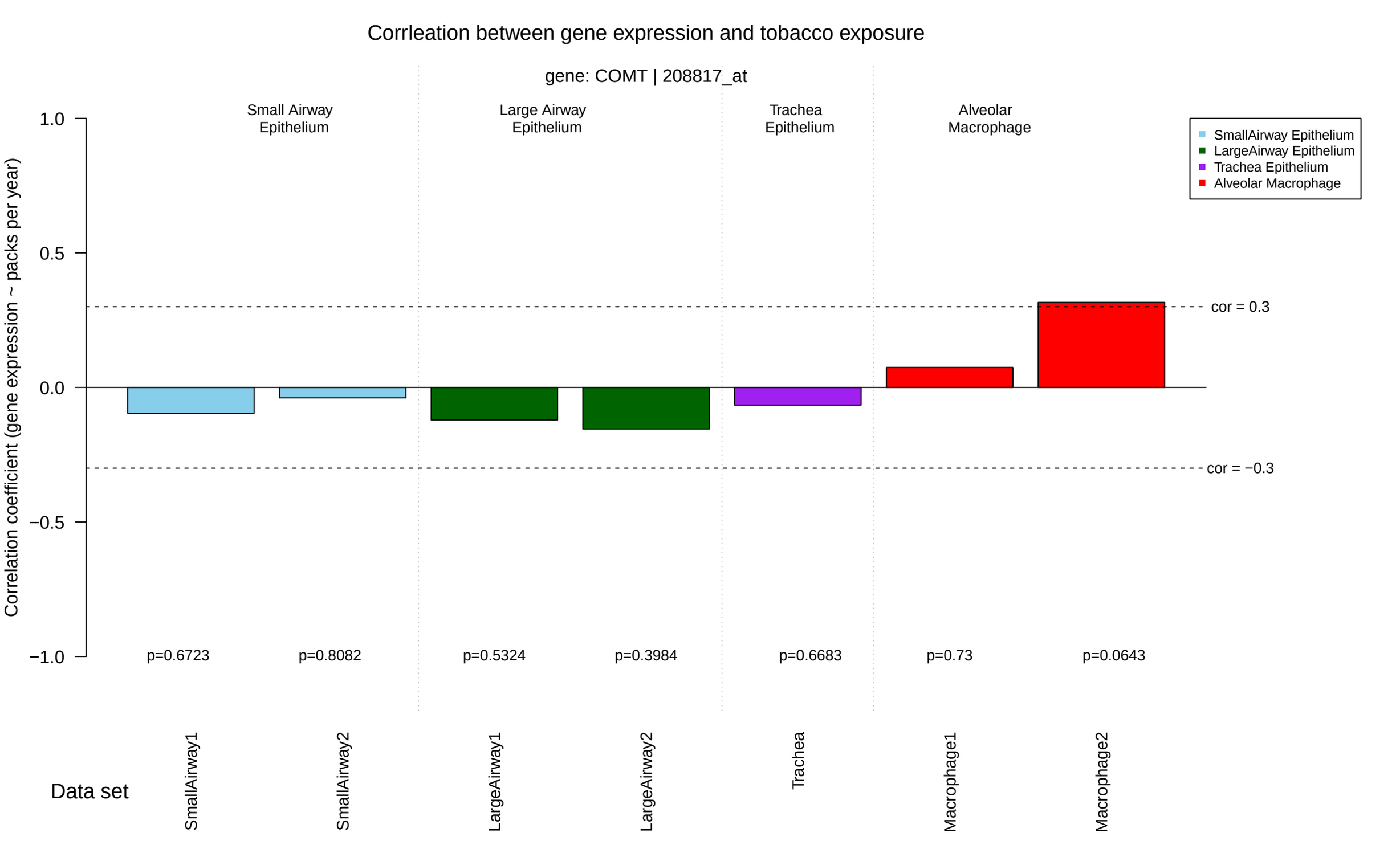

set: LargeAirway1 (p=0.0332)
COMT | 208817_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6543)
COMT | 208817_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5942)
COMT | 208817_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9474)
COMT | 208817_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.015)
COMT | 208817_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COMT | 208817_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6723
p=0.8082
p=0.5324
p=0.3984
p=0.6683
p=0.73
p=0.0643
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2